

Training: TCN_MHA (legacy)

How the previous production model was trained, from the legacy script.

TCN_MHA was not trained by this project. It is the previous production model, trained by the script `legacy/TCN_MHA.py` (also `legacy/TCN_MHA.ipynb`) on the old pipeline's h5 datasets, and loaded here only for comparison. Its training history was not saved, so only the script's settings can be documented.

Training settings in `legacy/TCN_MHA.py`

setting	value
data	<code>data/hrv_dataset.h5</code> (training) and <code>data/hrv_validation.h5</code> (validation), loaded entirely into RAM and copied to the GPU; built by the old preprocessing, whose subject split was not re-checked against this project's
inputs	the 30 RR of the window (one channel) + 12 static features: <code>rmssd</code> , <code>rs</code> , <code>ccm</code> , <code>ccm_n5</code> , <code>guzik</code> , <code>nn20</code> , <code>porta</code> , <code>ar_1..ar_5</code> (Burg AR(5) coefficients), as stored in the h5 files
target	<code>dRR[n+1]</code> as stored in the h5 files (no RLS residual)
loss	$\text{RSAPhaseAwareLoss} = \text{Huber}(\text{delta } 1) + 0.2 \times \text{mean}(\text{relu}(-y_{\text{true}} - y_{\text{pred}}))$ (a sign penalty) + $0.1 \times \text{var}(y_{\text{true}}) - \text{var}(y_{\text{pred}}) $ over the batch (a variance-matching penalty)
optimiser	AdamW, learning rate $5e-4$, weight decay $1e-4$
batch, epochs	batch 128 (chosen 'to preserve local gradient variance in the custom loss'); EPOCHS = 3 in the script as it is in the repository
callbacks	EarlyStopping (patience 12, restore best), ReduceLROnPlateau (factor 0.5, patience 5, min $1e-6$), ModelCheckpoint of the best <code>val_loss</code> to <code>tcn_attention_hrv_best.keras</code>
regularisation	SpatialDropout1D(0.1) in every residual block, weight decay
reproducibility	seed 211, deterministic TensorFlow ops; the saved weights are dated 2026-10-06 22:01

Differences from this project's training

- Its features are recomputed per window by `legacy/utlis2.py` (including a Burg AR fit), not by the streaming engine, so they cannot run beat by beat on the microcontroller.
- No RLS residual: it learns the whole dRR, not a correction of an adaptive forecast.
- The variance term of its loss rewards predictions that oscillate as much as the true dRR; with shuffled batches it mostly measures the spread between subjects. This project dropped it for plain Huber, and it is a plausible cause of the RSA peaks TCN_MHA invents in the closed loop.
- Evaluated here as a float model only (no int8 export): validation open loop 33.72 ms.

Abbreviations and acronyms used in this document

term	meaning
AdamW	Adam optimiser with decoupled weight decay (the training algorithm of the neural models)
AR	autoregressive: a value predicted as a weighted sum of past values; AR(p) uses p past values
ar	feature-set name: autoregressive inputs (rr_dev and the 8 most recent dRR) of the linear model
ccm	complex correlation measure: non-linearity index of the Poincare plot of consecutive RR intervals
dRR	beat-to-beat change of the RR interval: $dRR[n+1] = RR[n+1] - RR[n]$ (ms); the quantity predicted
GPU	graphics processing unit
guzik	feature: Guzik asymmetry: sum of positive dRR / sum of dRR
HRV	heart rate variability
int8	8-bit integer: weights and activations quantised to 8-bit integers, as run on the microcontroller; _int8 = the int8 version of a run
MHA	multi-head attention (a transformer layer, not supported by the MCU toolchain)
ms	milliseconds
nn20	number of successive RR differences larger than 20 ms in the window
porta	feature: Porta asymmetry: count of negative dRR / count of non-zero dRR
RAM	random-access memory
RLS	recursive least squares: an adaptive filter that updates its coefficients at every beat
rmssd	root mean square of successive differences of the RR intervals in the window
RR	RR interval: time between two consecutive heartbeats (R waves of the ECG), in ms
rs	rescaled range: (max - min of the cumulative deviations from the mean) / standard deviation
RSA	respiratory sinus arrhythmia: the oscillation of the heart rate driven by breathing
TCN	temporal convolutional network (causal, dilated 1-D convolutions)
tcn	temporal convolutional network (run-name part)
val	validation: the 16 validation subjects, used for model selection